

MECE E3610: Individual Term Project

Alfa Romeo P2 1924

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1 Requirements, Assumptions, and Part Breakdown

1.1 Project Overview

The purpose of this project is to design a scaled version of a 1920s era race car and prepare for manufacture. Below, I design, model, and develop a manufacturing plan for the **1924 Alfa Romeo P2** according to the project option **A**.



Figure 1: Alfa Romeo P2 1924.

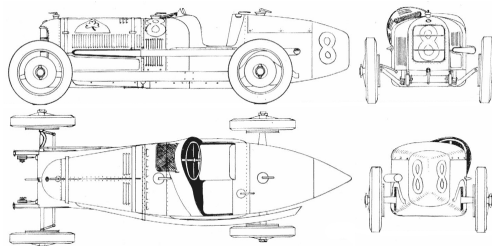


Figure 2: Alfa Romeo P2 1924 Orthographic Views.

During modeling, I made approximations around complex curves, especially those that served little aesthetic purpose and would make casting of the part harder. For instance, I used the approximation made in the blueprint for the shape of the seats in the cockpit instead of modeling the complex, tight curve seen in some top views. In addition, the cockpit pod was extruded with

a filleted shape, though not a rectangle because of the unique position of the wheel in this car. All wheels are approximated to be the same width and height.

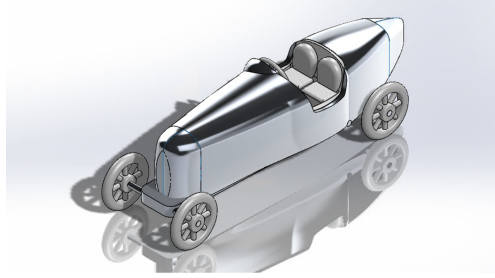


Figure 3: Toy Assembly Model

1.2 Model Specifications

Category	Specification / Assumption	Justification
Scale	1:24 Scale	To have a similar scale to the group project, 1:18 is in that range and is a common collector scale that still allows for high detail in the open cockpit.
Dimensions	Length ~ 6.5 in, Max Width: ~ 2.5 in.	Based on the original 155.5 in length scaled down.
Target Quantity	1	For Option A we are manufacturing a "one-off" product. Small-batch production, therefore we don't need permanent/ultra durable tooling.
Surface Finish	As cast, flash and parting line removed.	Much like in the Silicone application of the group project, the casting will have flash and a parting line. This is to be sanded off with a medium grit, so as to smooth the surface discontinuity off. The end product will still have some roughness (as cast).
Strength Req.	Total weight on axles ~ 1 lb.	Axle material chosen with yield strength of 40,000 psi. Wheels must be able to withstand the weight of the car body, chassis, and cockpit, thus ABS material selection. Ensures the thin axle rods do not sag under the car model's weight.
Safety	Al 356, 6061, PLA, ABS	Must comply with standard consumer toy safety regulations regarding sharp edges.
Draft Angle	Between 1 and 2 degrees	Will allow for ease in removal of casting from mold.

1.3 Parts and Process Selection

Part Name	Material	Process
Car Body	A356 Aluminum	Sand Casting
Chassis Frame	6061 Aluminum	Abrasive Water Jet
Wheels (4)	ABS Plastic	Injection Molding
Axles (2)	6061 Aluminum	Cut-to-length
Fastening Rod	6061 Aluminum	Cut-to-length
Cockpit	PLA	FDM 3D Printing

1.3.1 Car Body

- **A356 Aluminum:** A common, cost-efficient casting alloy. Known for excellent castability due to its higher ductility, fluidity and strength. It also has a lower thermal expansion coefficient than other common Aluminum Alloys, which is good for casting. Its fluidity will help with thin features.
- **Sand Casting:** Sand casting is highly cost-effective for a one-off scaled model because it eliminates expensive tooling, allowing us to use a 3D-printed FDM pattern and inexpensive sand. Paired with the high fluidity of Aluminum A356, this will ensure that the metal successfully fills the thin features towards the edges of the Alfa Romeo body. A negative side effect of sand casting is, however, the poor surface quality, as we will get a grainy finish at the end.



Figure 4: Car Body Pattern/Product

1.3.2 Chassis

- **6061 Aluminum Alloy:** Using a 0.25 inch thick 12x12 in plate (source example from SpeedyMetals). 6061 is considered strong enough for applications like this car, while being significantly more affordable than a “stronger” alloy like 2024, meaning the chassis will be able to support the weight of the car body, and by using a flat slab as the material, we can make the process quite fast and straightforward for a one-off manufacturing process.
- **Abrasive Water Jet:** While stamping is the industry standard for high-volume chassis production, I propose using the Abrasive Water jet process for this project. Since the production quantity is singular, the cost of creating stamping dies is not justifiable. Water jet cutting provides the required tolerance for axle alignment without any upfront tooling investment.

In addition, the cold-cutting process avoids any thermal distortion that could affect the mechanical properties for which we chose Aluminum 6061. Drill the Front, Rear axle holes with clearance fit. Drill the fastening rod with interference fit at 1.42 inches from the rear end of the chassis.

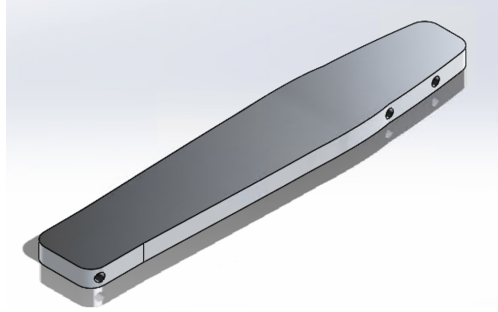


Figure 5: Chassis

1.3.3 Rods (Front, Rear, Fastening)

- **0.125" diameter 6061 Aluminum Alloy rods:** These rods are commonly available and used for small projects like this, a 6 inch length rod should be enough for both the fastener, front, and rear wheel tracks. They are heat treated for strength, and 6061 alloy, with a nominal yield strength of 40,000 psi will hold up against creep, so these rods should work well to support the car body.
- **Cut-to-length:** Saw to length (Front/Rear tracks ~ 2.1 in, Fastening ~ 1.47 in).



Figure 6: Stock Rod Example

1.3.4 Wheels

- **ABS:** To ensure the heavier metal components do not cause the wheel spokes to sag over time (creep), ABS (Acrylonitrile Butadiene Styrene) was chosen. This is a thermoplastic polymer with high rigidity and potential for polished/smooth surface finish.
- **Injection Molding:** Machine the mold from an Aluminum 6061 rectangular rod. First cut it to desired height, then proceed with the machining sequence. This will allow for good surface

quality of the ABS wheels. Center diameter of wheel should allow for line or interference fit with axles.

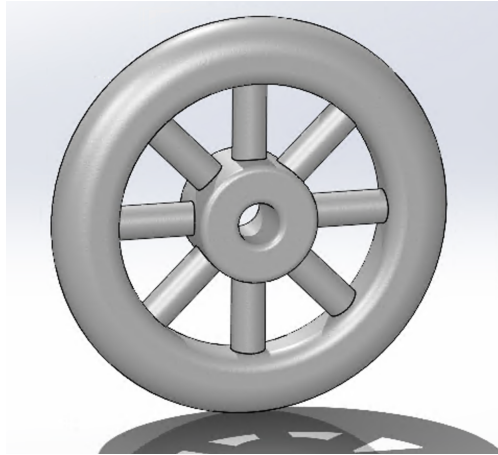


Figure 7: Wheel

1.3.5 Cockpit

- **PLA:** PLA is one of the most common 3D printing materials and is readily available in the Makerspace. The cockpit is an aesthetic component, so tolerances will not be a cause for concern.
- **3D Printing:** The intricate geometry of the cockpit design's features, such as the seat cushion and steering wheel would be quite difficult to produce using subtractive manufacturing like milling or casting manufacturing methods, which requires a parting line from which the casting could realistically be removed from the mold. Since additive manufacturing can create these detailed internal and external features without the need for complex tooling, FDM is the most effective and accessible process for this component.

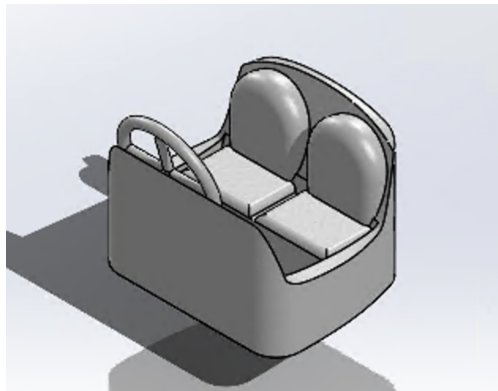


Figure 8: Cockpit

1.4 General Assembly Sequence

1. The cast body will be designed with 1 integrated boss that will secure the chassis to the car body.
2. Sand off flash and parting line from casting.
3. Drill holes in chassis (interference fit for fastening, clearance for axles).
4. Place the car body flush onto the chassis, aligning the corresponding holes.
5. Fasten the body to the chassis by inserting the cut-to-length securing rod. Insert the cut to length front and back axle rods as well.
6. Press-fit wheels onto cut-to-length axles.
7. Insert cockpit.

2 Sand Casting & Car Body Tooling Analysis

2.1 Sand Casting Calculations

Calculations for the car body sand casting procedure are shown below. An unpressurized gating system was used to limit excessive turbulence when pouring. For this application, I selected a gating ratio: $A_s : A_r : A_g = 1 : 2 : 2$, which is a medium between pressurized and unpressurized. This is in order to keep the gating area minimal in the finished casting, and to reduce the need for venting. In order to achieve “laminar-like” flow ($Re < 2000$) to minimize turbulence when pouring, a slightly longer pouring time of 5 seconds was assumed. This is permissible given the insulating characteristics of sand casting meaning it should not start solidifying within this pouring period.

Assumptions:

- A356 molten viscosity: $1.13 \times 10^{-3} kg/m/s$
- Filling time (t_f): 5s
- Pouring Height (h): 2in
- Discharge Coefficient (C_d): 0.5

2.1.1 Volumetric Flow Rate

Given Vol $V = 8.60 \text{ in}^3$:

$$Q = \frac{V}{t_f} = \frac{8.60}{5} = 1.72 \text{ in}^3/s$$

2.1.2 Gating System

Velocity at sprue base (v_s):

$$v_s = \sqrt{2gh} = \sqrt{2 \cdot 386.09 \cdot 2} \approx 39.3 \text{ in/s}$$

Sprue Choke Area (A_s):

$$A_s = \frac{Q}{C_d \cdot v_s} = \frac{1.72}{0.5 \cdot 39.3} \approx 0.0875 \text{ in}^2$$

Sprue diameter:

$$d_s = \sqrt{4 \cdot A_s / \pi} = 0.33 \text{ in}$$

The top diameter can be approximated as 1.5x A_s for our 2 in pour height.

$$A_{top} = 1.5(A_s) = 0.127 \text{ in}^2$$

$$A_g = 2(A_s) = 0.175 \text{ in}^2$$

$$A_r = 2(A_s) = 0.175 \text{ in}^2$$

For a rectangular runner, if we assume $2h = w$:

$$h_r = 0.31 \text{ in}$$

$$w_r = 0.60 \text{ in}$$

Ingate Geometry: Using two in-gates with rectangular cross sections along the parting line,

- Area per gate = $A_g/2$
- gate thickness typical values range: $t \approx 4 \text{ mm}$ (0.157 in), from Lecture.
- Gate width: $w_g = \frac{A_g}{2t} = 0.56 \text{ in}$
- Gate velocity: $v_g = \frac{Q}{A_g} = 9.83 \text{ in/s}$

2.1.3 Laminar Flow Verification

To verify the Reynolds number, we need to calculate the hydraulic diameter first:

$$D_h = \frac{(4)(w_g)(t)}{2(w_g + t)} = 0.245 \text{ in}$$

$$Re = \frac{v_g \cdot D_h}{\nu} = \frac{9.83 \cdot 0.245}{0.0017} \approx 1417$$

Since $1417 < 2000$, the flow is laminar.

2.1.4 Venting Requirements

The required vent area is equivalent to the gating area, since the total metal volume in is the total air volume out. My proposal for the vents consists of three cylindrical vents:

$$r = \sqrt{\frac{A_g}{3\pi}} = 0.136 \text{ in}$$

$$\phi = 0.27 \text{ in} \times 3$$

2.2 Mold Design

Given the unique shape of the windshield in the car body, the 3D printed pattern will have to be divided into two parts. The general car body, and the part that protrudes over the hood as a small extension. A simple horizontal split on the body cad at this point would be sufficient. To keep the part from moving, a small sliding attachment could be used, with the male part on the small patch section. Once the mold is ready, the larger body pattern can be removed as normal, and then the patched section can be pulled out carefully horizontally, manually or with tweezers. This will not cause an issue in removing the cast body, given that the sand casting is disposable.

To accommodate the scaled body (6.5 x 2 in), the mold box was chosen to be 8.5 x 3 x 3. A small pouring basin was included with the sprue to account for the 2 inch pouring height chosen. The mold should have a draft angle around 2 degrees for removal. As per the calculations, the sprue has diameter 0.33", down to a runner with $h = 0.3$ ", $w = 0.59$ ". The gating system consists of two gates of thickness 0.157" (4mm) and width 0.56". Three cylindrical vents with diameter 0.27" at the top of the parting line account for the air volume exiting the mold.

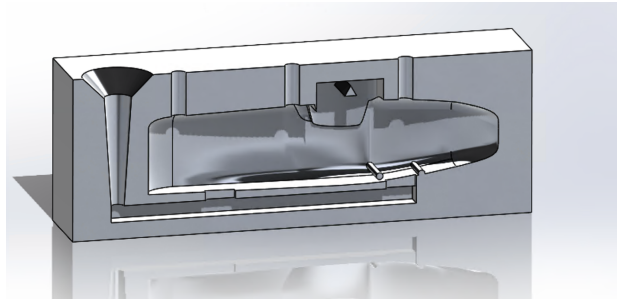


Figure 9: Car Mold CAD. Note, no standard sand material in Solidworks. Disregard material appearance.

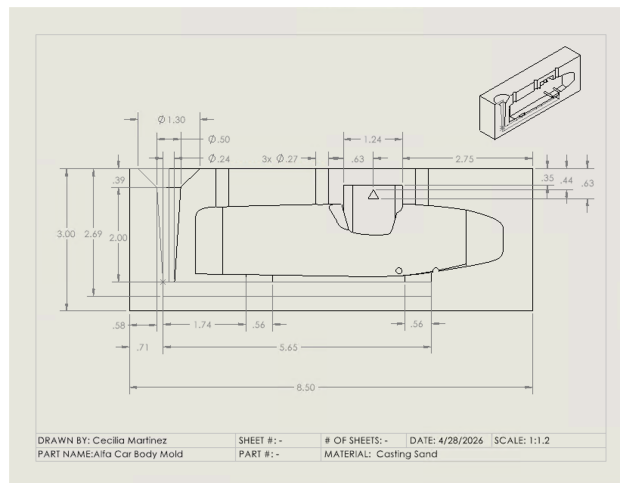


Figure 10: Car Mold drawing. Note, important dimensions included for clarity, though not fully dimensioned.

3 Wheel Mold for Injection Molding & Machining Plan

3.1 Wheel Mold Tooling Design

Borrowing from the sprue and vent sizing in the CAM tutorial video, a mold was designed for the wheel described in Milestone 1. The wheel will have a total diameter of 1.25 in and total thickness of 0.25 in, with 8 spokes. Below is a drawing of half of the mold. Note, the drawing below uses nominal dimensions, so it does not account for the shrinkage calculations below. It is not fully dimensioned, though a variety of key measurements are shown for clarity. No fillets under 1/37th inch were used. The half mold box is 1.875 x 1.875 x 0.5. Two alignment holes are included to ensure both mold-halves are properly aligned throughout injection and cooling.

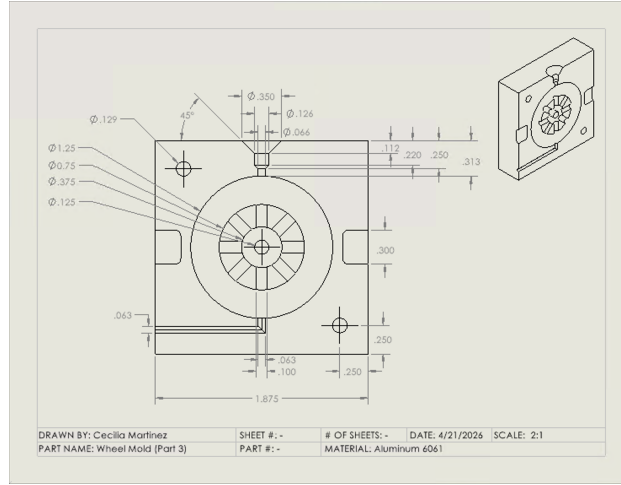


Figure 11: Drawing of half-cavity wheel mold for Alfa Romeo P2 1924 Toy Car.

A flat parting plane located at the wheel's mid-plane was chosen for this mold, as this is the most efficient choice for a symmetrical part like a wheel, allowing for straightforward machining of the cavity and ensures the part can easily be removed from the metal mold. For the gate, I maintained an edge gate feeding the rim of the wheel. The small cosmetic blemish from the gate left along the parting line during trimming will not be too noticeable and will not disrupt the wheel's function. Lastly, the ejection method is manual de-molding, with added ease via pry slots to open with a flat head screwdriver. For this one-shot iteration, manual de-molding is acceptable. The pry cavities allow for the mechanical removal of the part without damaging the wheel's geometry with unnecessary torsion.

3.2 Core Calculations for Injection Molding

3.3 Shrinkage Compensation

ABS shrinkage typically ranges from 0.7% to 1.6%. We have adopted a median shrink factor of 1.15% (0.0115) for our mold design to ensure the final part stays within tolerance. We compensate by oversizing the cavity and undersizing the core pin (for the axle hole).

$$D_{mold} = \frac{D_{part}}{1 - \text{shrink factor}} \quad (1)$$

- **Wheel Outer Diameter (OD):** To reach the target 1.25", we must oversize the outer diameter.

$$D_{cavity} = \frac{1.25 \text{ in}}{1 - 0.0115} \approx 1.264 \text{ in}$$

- **Axle Hole Inner Diameter (ID):** To reach the target 0.125", we must undersize the inner diameter.

$$D_{core} = \frac{0.125 \text{ in}}{1 + 0.0115} \approx 0.124 \text{ in}$$

3.3.1 Cooling Time Estimation

For the initial estimation, the cooling time is calculated using the simplified heat transfer equation:

$$t_{cool} \approx C \cdot h_{max}^2 \quad (2)$$

Where $h_{max} = 0.25 \text{ in} \approx 6.35 \text{ mm}$ and the empirical constant $C \approx 0.11 \text{ s/mm}^2$ for ABS. While this will give an approximate magnitude for the cooling time, using the empirical constant implicitly simplifies our model,

$$t_{cool} \approx 0.11 \cdot (6.35)^2 \approx 4.44 \text{ seconds} \quad (3)$$

The cooling constant $C \approx 0.11 \text{ s/mm}^2$ is an empirical coefficient derived from the one-dimensional heat conduction equation:

$$t_{cool} = \frac{h^2}{\pi^2 \alpha} \cdot \ln \left(\frac{4}{\pi} \cdot \frac{T_{melt} - T_{mold}}{T_{eject} - T_{mold}} \right)$$

It accounts for the thermal diffusivity ($\alpha = 0.9 - 0.13 \text{ mm}^2/\text{s}$) of ABS and the thermodynamic processing boundary conditions:

- **Melt Temperature (T_{melt}):** 220°C
- **Mold Temperature (T_{mold}):** 80°C
- **Ejection Temperature (T_{eject}):** 100°C (Glass transition temperature of ABS)

This constant serves as a "lumped parameter" that simplifies the complex logarithmic temperature ratio and thermal diffusivity of the polymer into a design coefficient suitable for rapid cycle time estimation.

3.4 CAM/Machining Plan

Since the square 1 7/8" mold is not an easily accessible stock size bar, the most efficient stock size for this mold is a 1/2" x 2" Aluminum Rectangle Bar 6061-T6511-Extruded, available at OnlineMetals.com. The first step in our sequence will be to cut down the stock size to 1/2" x 1.875" x 1.875". Then the mold can be machined in a CNC mill:

In order to do the facing in one step, chose a mill to fit the entire stock. A 2" mill is the most common type of face mill larger than our 1.875 in mold. An option would be the 45° Lead Angle, 1.846" to 2.205" Cut Dia, Indexable & Chamfer End Mill by Tungaloy. The second operation, spot drilling, creates the initial cavity for the alignment holes. An option for the twist drilling would be the Jobber Length Drill Bit: 1/8" (0.1250) Dia, 118 degrees, Solid Carbide from Precision Twist Drill. This step will finish making the pin holes up to the desired 0.129 in diameter with precision so the two mold parts align. The fourth operation, roughing, removes the "bulk" of the aluminum for the wheel cavity. Then, the smaller 1/8" mill will pass over the rough surfaces created to shave

Table 1: Machining Sequence and Tooling

Step	Operation	Tool	Tool Size
1	Face Mill Top	Face Mill	2"
2	Spot Drill Pins	Carbide Spot Drill	1/8"
3	Drill Alignment Holes	Twist Drill	1/8"
4	Roughing	4-flute Flat Endmill	1/4"
5	Finishing the Walls & Floor	4-flute Flat Endmill	1/8"
6	Finish Contour	Ball Nose Endmill	3/32"

off the jagged edges. These 4-flute Flat Endmills are commonly produced and available by Hass Tooling. The last operation is finishing; we smooth out the sharp corners with the specified fillets. Since I have "corners" with radii $R = 0.05"$, the ideal tool size will be $3/32"$, which has a diameter slightly less than $0.10"$.

3.5 Cutting Analysis (Roughing Operation)

The following parameters are based on machining Aluminum 6061 using a $1/4"$ 3-flute carbide end mill, the tool used in the roughing operation. This is the operation that will require the most force and involve the highest material removal per unit time.

3.5.1 Cutting Parameters

- **Cutting Speed (V_c):** 100 SFM (Surface Feet per Minute). This is a conservative approximation, given the large RPM range available in the Tormach PCNC 770, aimed at maintaining a safe and steady machining process.

$$SFM = \frac{RPM \times \pi \times D}{12},$$

so this about $\sim 1,528$ RPM.

- **Feed per Tooth (f_t):** 0.002 in/tooth. This is a standard starting recommendation for a $1/4"$ endmill on aluminum.
- **Depth of Cut (a_p):** 0.050 in. A general estimate in milling is to set the axial depth of cut between 20% and 50% of the tool diameter. With a $0.250"$ diameter tool, $0.050"$ is 20% of the tool diameter. This depth is chosen to keep cutting forces low, preventing excess tool deflection and vibration. It ensures that the mold cavity is roughed out efficiently without stressing the machine spindle or risking tool breakage.
- **Radial Width (w):** 0.125 in (50% stepover). A 50% radial stepover is a standard industrial practice for general-purpose milling. It provides a balanced load on the cutter, preventing the tool from being "buried" in the material.
- **Wheel Volume:** Relevant to material removal. Volume is 0.13 in^3 according to the mass properties feature in SOLIDWORKS.

3.5.2 Machining Calculations

- **Spindle Speed:** $RPM = \frac{V_c \cdot 12}{\pi \cdot D} \approx 1,528$ RPM
- **Feed Rate:** $F = RPM \cdot N_{flutes} \cdot f_t \approx 9.17$ IPM
- **Material Removal Rate (MRR):** $MRR = w \cdot a_p \cdot F \approx \mathbf{0.058}$ in³/min
- **Power Required (P):** $P = MRR \cdot 0.5 \approx \mathbf{0.029}$ HP
- **Cutting Force (F_c):** $F_c = \frac{P \cdot 33,000}{V_c} = \frac{0.029 \cdot 33,000}{100} \approx \mathbf{9.57}$ lbs

3.5.3 Machining Time Estimation

Considerations when approximating time to machine are as follows:

- **Conservative Material Removal Rate (MRR):** As justified in our parameter selection, a cutting speed of 100 SFM was utilized to ensure surface integrity and tool longevity. While this increases cycle time compared to high-speed machining, it significantly reduces the risk of tool deflection and catastrophic breakage, which is critical for a single-shot prototype mold.
- **Setup and Calibration:** Approximately 15 minutes is allocated to non-cutting activities, including part orientation, probing of the work coordinate system, and manual tool changes. These are necessary overheads for the Tormach PCNC 770 in a prototyping environment to ensure dimensional accuracy.

The roughing operation time is calculated based on the total volume of the cavity ($V = 0.13$ in³) and the Material Removal Rate ($MRR = 0.058$ in³/min). To account for non-cutting motions (rapid positioning, entry/exit), a safety factor (an efficiency consideration) (SF) of 1.8 is applied.

$$t_{roughing} = \frac{V}{MRR} \cdot SF \quad (4)$$

$$t_{roughing} = \frac{0.13 \text{ in}^3}{0.058 \text{ in}^3/\text{min}} \approx 2.24 \text{ minutes} \times 1.8 \approx 4.0 \text{ minutes} \quad (5)$$

Table 2: Estimated Time Breakdown of Machining Sequence (Single-Cavity Mold)

Operation	Tool Used	Time (min)	Justification
Setup & Probing	N/A	10	Manual orientation and setting.
Facing	Face Mill	1	Single pass to establish z-datum.
Roughing	1/4" Flat Endmill	4	Calculated based on 0.13 in ³ cavity volume.
Drilling	Twist Drill	2	Pilot/finish hole for alignment.
Finishing	1/8" Flat Endmill	10	Light passes for surface finish.
Tool Changes	Various	5	5 tools $\times$ 1 min per change.
Total		32	Total estimated time per mold half.

Finishing with smaller tools, slower feed rate and MRR will cause this process to be slower than roughing operations. In addition, finishing is expected to take a slightly longer because of the amount of fillets and the spoke shapes.

3.6 Machine Selection

The selected machine is the **Tormach PCNC 770**. With a 1 HP spindle, it significantly exceeds the 0.03 HP requirement, ensuring stability and accuracy for the mold cavity production. It is a compact CNC milling machine with cast-iron construction, 3-axis control, and high-speed spindle optimized for prototyping and small-part machining available at the teaching lab.

4 Injection Molding System Configuration & DFM Analysis

4.1 Mold System Configuration

Upon further analysis of the mold design in Part 3, the finalized configuration is as follows. For the small-scale manufacturing of the Alfa Romeo toy car, we only need to produce a total of four wheels. Thus, a single-cavity, two-plate mold is selected. This is the most efficient configuration for this part geometry, minimizing the number of moving parts and ensuring simplified maintenance. In this case, the runner system and the wheel part will be ejected together.

While producing the 4 wheel parts in a single shot would be more efficient in large scale production, the time savings do not compensate for the trade-offs in a one-off manufacturing process. The stock material would increase the capital cost, the larger piece would need higher clamping force, and higher shot volume allowance in the machine chosen. Moreover, choosing a single cavity mold, given the prototyping nature of this design, reduces initial tooling complexity and risk, allowing for easier iteration.

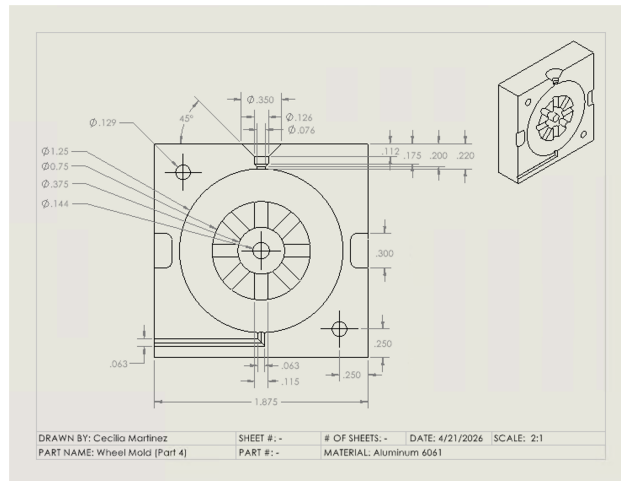


Figure 12: Drawing of half-cavity wheel mold for Alfa Romeo P2 1924 Toy Car. Note, this rendering reflects the shrinkage considerations for ABS plastic (1.15%), assuming the oversized outer diameter to be 1.264 in and the undersized pin diameter to be 0.124 in. Appropriate changes were made to the placement of other components such as the pry cavities and runner. Note, above drawing is not fully dimensioned to maintain clarity, though key dimensions were included. All fillets 0.05" unless otherwise noted.

4.2 Mold Flow, Ejection & Thermal System Design

A flat parting plane located at the wheel's mid-plane was chosen for this mold, as this is the most efficient choice for a symmetrical part like a wheel, allowing for straightforward machining of the

cavity and ensures the part can easily be removed from the metal mold. For the gate, I maintained an edge gate feeding the rim of the wheel. The small cosmetic blemish from the gate left along the parting line during trimming will not be too noticeable and will not disrupt the wheel's function. In addition, the edge gate is aligned with one of the spoke directions, this is intended to allow for faster fill time, under the assumption that the introduced turbulence is negligible.

4.2.1 Ejection System

The mold design includes two pry-cavities at the edge of the mold. However, to ensure the mold can be properly opened, the proposed system involves 2 ejector pins at the two corners where the alignment pins are.

- **Configuration:** Two ejector pins with a diameter of 0.129 in, the same diameter as the alignment pin holes.
- **Placement:** As we discussed in the design of the mold, the alignment pin locations on the mold are just holes on the half-cavity. We can exploit this design by modifying one of the half-cavity mold parts to have blind holes for this part, and maintain the through holes on the other half. The resulting interface will leave the 0.129 in ejector pins exposed on one side, such that a couple taps with a hammer, assisted by a screwdriver for pressure if necessary, will release the mold.

4.2.2 Cooling System

The thermal management system is designed to maintain a consistent mold temperature, which is critical for ensuring the dimensional accuracy of the ABS part and preventing warpage.

- **Layout:** Cooling channels should be strategically placed to remove heat efficiently.
- **Purpose:** The circulation of coolant fluid through these channels allows for active heat extraction. This maintains a steady state for the mold temperature, ensuring that the 1.15% shrinkage compensation remains effective across all produced units and reducing the overall cycle time by optimizing the cooling phase.

4.3 Design for Moldability (DFM) Analysis

4.3.1 Draft Angles

ABS standards for injection molding suggest a draft angle between 1 and 2° for removal. Given that the machined mold is not polished, the surface quality of the mold necessitates a draft angle of 2° to be applied to all walls perpendicular to the direction of removal, meaning all vertical walls. This is essential to prevent part binding due to friction against the mold walls during ejection. Because of the geometry of the wheel, this will be applied at the parting line for the spokes, given the tangent surface is vertical at the parting line. The same applied to the wheel's outer diameter. Lastly, the draft angle must be applied to the pin diameter, since this surface is also perpendicular to the removal direction.

4.3.2 Shrinkage Compensation

Since we initially selected ABS as the wheel material, our shrinkage calculations remain the same during the injection molding considerations. To reiterate, ABS shrinkage typically ranges from 0.7%

to 1.6%, so we have adopted a median shrink factor of 1.15% (0.0115) for our mold design to ensure the final part stays within tolerance. We compensate by oversizing the cavity and undersizing the core pin (for the axle hole), meaning that our wheel cavity's outer diameter is now 1.264" and the pin hole diameter is 0.124". Calculations are explicitly shown in section 3.3. This is accounted for in the mold design by using the scale feature on the wheel with 1.15% on SOLIDWORKS to ensure all proportions remain the same after cooling. See for the updated mold configuration.

4.4 Core Calculations (Process Parameters)

4.4.1 Shot Volume and Mass

The total shot volume includes the part geometry and the runner system. Based on the CAD-derived volume of 0.13 in³ for the wheel component and the delivery system dimensions specified in the tool drawing, we calculate the process parameters. The total shot volume (V_{shot}) is the sum of the part volume and the delivery system volume (sprue, runner, and gate). The part volume was derived from CAD mass properties, while the delivery system was calculated using the geometric dimensions provided in the tool layout.

1. **Part Volume** (V_{part}): 0.1300 in³

2. **Tapered Sprue** (V_{sprue}): ($\varnothing 0.350$ to $\varnothing 0.126$ over 0.114 depth):

$$V_{sprue} = \frac{1}{3}\pi(0.114)(0.175^2 + 0.175 \cdot 0.063 + 0.063^2) \approx 0.0054 \text{ in}^3 \quad (6)$$

3. **Runner and Gate** ($V_{run/gate}$):

$$V_{run/gate} \approx 0.0014 \text{ in}^3 \quad (7)$$

Total Shot Volume: Given the density of ABS ($\rho \approx 0.038 \text{ lb/in}^3$).

$$V_{shot} = 0.1300 + 0.0054 + 0.0014 = 0.1368 \text{ in}^3 \quad (8)$$

Given the material density, the total shot mass (m_{shot}) is:

$$m_{shot} = V_{shot} \cdot \rho = 0.1368 \text{ in}^3 \cdot 0.038 \text{ lb/in}^3 \approx 0.0052 \text{ lbs} \quad (9)$$

4.4.2 Clamping Force

The clamping force is determined by the projected area of the part and the injection pressure. The projected area (A_p) is the silhouette of the 1.25" diameter wheel on the parting plane.

$$A_p = \pi \cdot r^2 = \pi \cdot (0.625 \text{ in})^2 \approx 1.23 \text{ in}^2 \quad (10)$$

Standard pressure recommendations for injection molding of ABS suggest a range of 50-70 MPa, this is about 7,000 - 15,000 psi. Assuming a reasonable injection pressure (P) of 10,000 psi:

$$F_{clamp} = A_p \cdot P = 1.23 \text{ in}^2 \cdot 10,000 \text{ psi} = 12,300 \text{ lbs} \approx 6.15 \text{ tons} \quad (11)$$

4.4.3 Thermal Cooling Analysis

To ensure part quality and prevent post-molding deformation, we should be more precise when estimating the cooling time. This will be the largest time interval during the injection molding process. Cooling time was derived from first principles using the 1D heat conduction model for an infinite slab.

Given the provided material properties for ABS:

- Thermal Conductivity (k): 0.175 W/m·K
- Density (ρ): 1040 kg/m³
- Specific Heat (C_p): 1850 J/kg·K

The thermal diffusivity (α) is calculated as:

$$\alpha = \frac{k}{\rho \cdot C_p} = \frac{0.175}{1040 \cdot 1850} \approx 0.0909 \text{ mm}^2/\text{s} \quad (12)$$

The half-thickness ($h = 0.125 \text{ in} = 3.175 \text{ mm}$) is utilized in the Fourier conduction model as the characteristic distance to account for the symmetrical heat dissipation. This represents the maximum distance heat must travel from the part's thermal center to the mold interface. Using the Fourier heat conduction model, the cooling time (t_c) required to reach the ejection temperature ($T_{eject} = 90^\circ\text{C}$) from the melt temperature ($T_{melt} = 215^\circ\text{C}$) with a mold temperature ($T_{mold} = 70^\circ\text{C}$) is:

$$t_c = \frac{h^2}{\pi^2 \alpha} \ln \left(\frac{4 T_{melt} - T_{mold}}{\pi T_{eject} - T_{mold}} \right) \approx 25.0 \text{ seconds} \quad (13)$$

4.5 Machine Selection & Cycle Time Estimation

4.5.1 Clamping Force Determination

To determine the required machine tonnage, the clamping force must exceed the pressure exerted by the molten ABS during the injection phase.

- **Projected Area (A_p):** The circular wheel has a diameter of 1.25", resulting in a projected area of:

$$A_p = \pi \cdot r^2 = \pi \cdot (0.625 \text{ in})^2 \approx 1.23 \text{ in}^2 \quad (14)$$

- **Required Clamping Force (F_c):** Much like the campling force, we assuming an injection pressure (P_{inj}) of 10,000 psi. However, to be safe in our machine sizing, a standard safety factor (S) of 10% is used to prevent parting line flash:

$$F_c = A_p \cdot P_{inj} \cdot S = 1.23 \text{ in}^2 \cdot 10,000 \text{ psi} \cdot 1.1 = 13,530 \text{ lbs} \quad (15)$$

Converting to tonnage (where 1 ton = 2,000 lbs):

$$\text{Tonnage} = \frac{13,530 \text{ lbs}}{2,000 \text{ lbs/ton}} \approx 6.77 \text{ tons} \quad (16)$$

Given this requirement of 6.77 tons, a machine with a minimum 7-ton capacity is recommended.

4.6 Machine Selection & Cycle Time Estimation

4.6.1 Machine Selection: Babyplast 6/12 Standard

The Babyplast 6/12 Standard (Horizontal) injection molding machine was selected for this project. As the industry-standard configuration, this horizontal press provides a robust, proven platform for single-cavity prototype production.

- **Tonnage Capacity:** The machine provides a clamping force of 62.5 kN (≈ 7.02 US tons). This provides an ideal safety margin over our calculated requirement of 6.77 tons (at 10,000 psi), ensuring the mold remains securely clamped throughout the injection cycle.
- **Trade-off:** While all other considerations for the mold work well, the mold design is 1" thick, meaning it does not meet the 70 mm (2.75") requirement. Nevertheless, this can be resolved with backing plates, so it will not be an issue.

4.6.2 Cycle Time Estimation

The cycle time is derived from the machine's official dry cycle specifications and the calculated thermal requirements for the ABS material. By using the manufacturer's specified dry cycle of 2.4s, we establish a precise mechanical limit for the production process.

4.6.3 Total Cycle Time Estimation

The estimated cycle time combines the mechanical machine limits with the material-dependent cooling requirement.

Table 3: Optimized Cycle Time (Babyplast 6/12 Standard Horizontal)

Phase	Time (s)	Justification
Fill	1	Controlled injection for small-volume ABS is often about 1 s or less.
Pack	2.5	Packing phase to ensure part density.
Cooling	25.0	Calculated via Fourier heat conduction model.
Dry Cycle	2.4	Manufacturer-specified mechanical limit.
Total Cycle	30.9	Total cycle time per part.

Cooling accounts for 79.6% of the total cycle time. This analysis indicates that the efficiency of the mold's cooling channels is the critical success factor for the project, necessitating active thermal management via the Babyplast's 120 l/h cooling water loop. Of course, set-up and removal will add to the practical cycle time, though the ejector pins should speed up the process.

5 Abrasive Water Jet Cutting of Chassis

Water jet machining was chosen for the production of the chassis due to its ability to produce precise cuts without the risk of introducing heat-affected zones, which can distort the aluminum or affect the lifespan of the weight-bearing part. To effectively cut the Aluminum 6061, the water jet system will be operated at a pressure of 55,000 PSI, as this will provide the best quality cuts.

5.1 Constants

- Discharge Coefficient (C_d): 0.60
- Orifice Diameter (d): 0.015 in
- Pressure (ΔP): 55,000 PSI
- Water Density (ρ): 0.0361 lb/in³

6 Calculations

6.1 Perimeter and Time

Given dimensions $L = 6.5$ in and $W = 1.5$ in:

$$P = 2(L + W) = 2(6.5 + 1.5) = 16.0 \text{ in}$$
$$t = \frac{P}{v} = \frac{16.0 \text{ in}}{1.0 \text{ in/s}} = 16.0 \text{ s}$$

6.2 Volumetric Flow Rate (Q)

First, calculate the orifice area (A):

$$A = \frac{\pi d^2}{4} = \frac{\pi(0.015 \text{ in})^2}{4} \approx 1.767 \times 10^{-4} \text{ in}^2$$

Using the Bernoulli-derived orifice equation:

$$Q = C_d A \sqrt{\frac{2\Delta P}{\rho_{mass}}}$$

Where $\rho_{mass} = \frac{0.0361 \text{ lb/in}^3}{386.4 \text{ in/s}^2} \approx 9.34 \times 10^{-5} \text{ lb}_f \cdot \text{s}^2/\text{in}^4$:

$$Q = 0.60 \times (1.767 \times 10^{-4}) \times \sqrt{\frac{2(55,000)}{9.34 \times 10^{-5}}}$$
$$Q \approx (1.06 \times 10^{-4}) \times 34,318$$
$$Q \approx 3.64 \text{ in}^3/\text{s}$$

Note, 3.64 in³/s is approximately 0.95 gallons per minute (GPM), which is a standard flow rate for a 0.015" orifice at this pressure. Once the chassis shape is cut, the holes for the axle and fastening rods must be tapped. Then, the part is complete. This concludes the major assembly component analyses for the Alfa Romeo Toy Car.

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AI Use Statement

Google Gemini was used to help structure the references section in LaTeX after encountering some compilation issues.